NASH Dhabha & Bakes: Masala Mitra

Proposal: a voice-enabled RAG chatbot waiter

End-Semester Project · Review – I (Phase 1)

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25ARE471 Introduction to Generative AI · Guided by Prof. Rakesh Kumar

Agenda

1. Course context and project fit
2. Motivation and pain points
3. Problem definition and objectives
4. Users and existing approaches
5. Proposed solution, AI-waiter plan, RAG architecture
6. GenAI concepts and planned tools
7. Knowledge-base design
8. Phase-1 prototype
9. Planned evaluation
10. Work plan, risks and course outcomes

Course Context: Where This Project Fits

Unit 1
Introduction to GenAI Technology

- ✓ LLM operations: tokenisation, embeddings
- 📖 Traditional AI vs generative AI
- ✓ Responsible AI: bias, fairness, explainability
- ✓ Guidelines on GenAI usage

Unit 2
GenAI Tools & Applications

- ✓ Prompt engineering: role-task-output, few-shot
- ✓ RAG: vector embeddings, semantic & similarity search
- ✓ Text processing: Q&A, correction
- 📖 Basics of fine-tuning

Unit 3
Advances in GenAI

- 📖 AI agents and agentic flow
- ✓ AI-augmented workflow: coding assistance
- ✓ LLMOps concepts
- ✓ Human-AI interaction (HAX)

✓ = topic applied in this project (from the 25ARE471 syllabus)

Motivation / Pain Points

At a busy dhaba, the menu has the answer, but nobody has time to explain it.

Waiters are stretched

At peak hours, guests wait just to ask a question or place an order.

Menus can't answer questions

Is it spicy? Veg? Dry or gravy? Under ₹250? A printed or QR menu can't tell you.

Special requests get lost

"Less oil, no onion" said at the table often never reaches the kitchen.

People ask in their own way

Hinglish, chat-speak and Indian spellings ("parota", "briyani") confuse simple bots.

Problem Definition / Objectives

PROBLEM

Build a waiter-like assistant that understands natural, mixed-language requests (text or voice), answers only from the restaurant's real menu, and handles the order from suggestion to kitchen, bill and payment.

OBJECTIVES

- **O1 · Understand:** mood, diet, spice, dry/gravy, price, chat-speak
- **O2 · Ground:** retrieve from a menu knowledge base; no invented dishes or prices
- **O3 · Voice:** speech-to-text chat and voice notes for the chef
- **O4 · Complete the loop:** cart → chef notes → kitchen → tracking → bill → payment → feedback
- **O5 · Evaluate:** define metrics and test sets before building, measure in Phase 2

Users & Scenarios



Customer

- Asks: "something spicy and veg under 250"
- Speaks instead of typing
- Wants live order status and the bill in one place



Chef / kitchen

- Needs every special request, clearly
- Can play voice notes on the ticket
- Sees dine-in table or takeaway



Owner

- Fewer missed orders and requests
- Feedback collected after every meal
- Runs on a laptop; no paid AI needed

Existing Approaches & the Gap

Capability	Printed menu	QR menu	Generic chatbot	Masala Mitra (proposed)
Answers questions about dishes	✗	✗	✓	✓
Answers only from the real menu	✓	✓	✗	✓
Understands Hinglish & chat-speak	✗	✗	~	✓
Voice input + voice chef notes	✗	✗	~	✓
Order, kitchen, bill, payment in one flow	✗	~	✗	✓
Mood-based suggestions	✗	✗	~	✓

✓ yes ~ partly ✗ no

Proposed Solution: Masala Mitra

Greets & asks your mood

Suggests dishes that fit

Answers like a waiter

From the real menu only (RAG)

Voice chat

Speech-to-text in the browser

Voice chef notes

Recorded and sent to the kitchen

Menu with filters

Veg / non-veg, search, sections

Kitchen display

Every ticket with notes

Live order tracking

Sent to chef → cooking → ready

Bill & payment in chat

GST bill, UPI / card / cash

Planned AI-Waiter Interactions

A **Greeting & general**

Hello, what can you do, where to sit

B **Menu & food**

Today's special, spicy or sweet, price, Jain

C **Orders**

Order, track, how long, change or cancel

D **Assistance**

Call the waiter, water, chair, washroom

E **Needs & complaints**

Allergies, no spicy, cold food, urgent help

F **Payment & feedback**

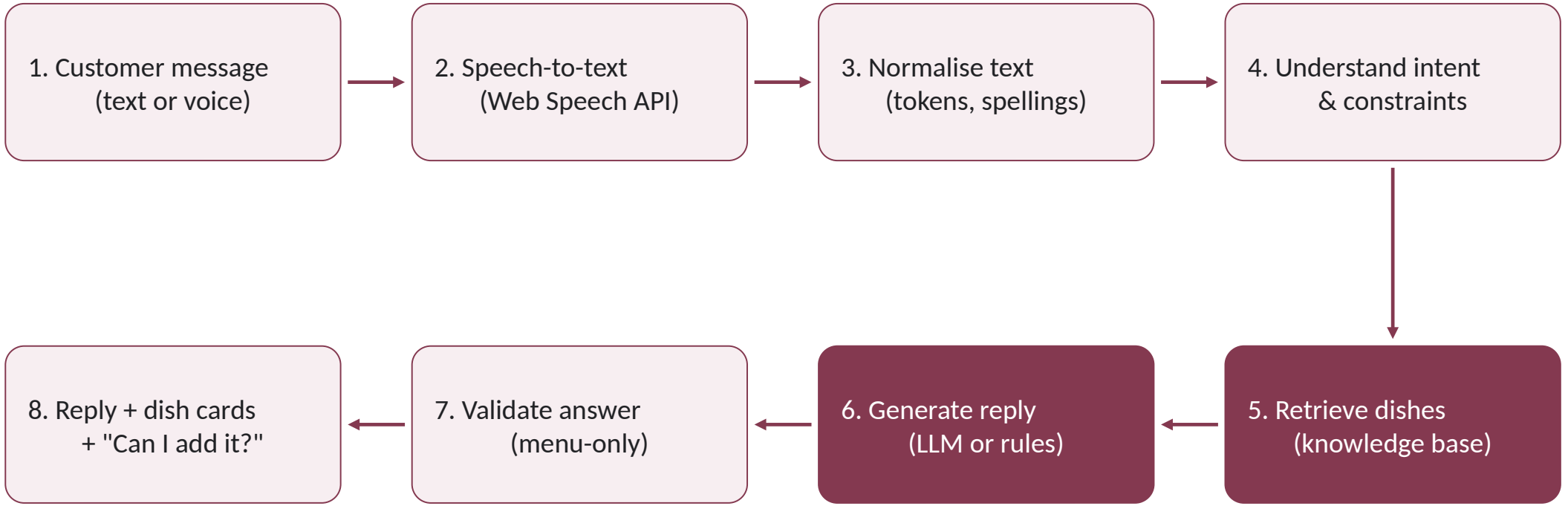
UPI, bill, split bill, feedback

Design rules

- Never invent ingredients, allergens, prices or order status
- Not sure? Hand over to a staff member
- English, Hindi and Tamil, by text or voice
- Staff desk with requests and satisfaction

Phase 2: every group becomes a saved test set, with held-out questions

Proposed RAG Architecture



Plan: the LLM writes only from retrieved menu facts, and a validator checks every answer.

GenAI Concepts Applied (from the Syllabus)

Tokenisation

Split and normalise messages: stop-words, stemming, chat-speak mapping

Vector representations

TF-IDF vectors for dishes and house info (embedding-style)

Similarity search

Cosine similarity ranks dishes for each question

RAG

Generate answers only from retrieved menu context

Prompt engineering

Role · Task · Output-format prompt with a JSON schema

Responsible AI

Grounding, diet safety, honest "not on menu" replies

Planned Tools & Tech Stack

Layer	Tool	Purpose	Cost
Retrieval	TF-IDF + structured dish facts (JavaScript)	Find the right dishes and house info	Free
Generation	Google Gemini (free tier) + rule-based writer	Write waiter replies from retrieved facts	Free tier
Speech	Web Speech API · MediaRecorder	Voice chat · voice chef notes	Free (browser)
Backend	Node.js + Express, JSON store	Sessions, orders, voice files, kitchen	Free
Frontend	HTML, CSS, vanilla JavaScript	Website, chat, menu, checkout	Free
Testing	Playwright (real browser)	Automated end-to-end evaluation	Free
Dev workflow	Claude Code (AI coding assistant)	Faster build, tests and fixes	—

Knowledge-Base Design

Menu sections (10)

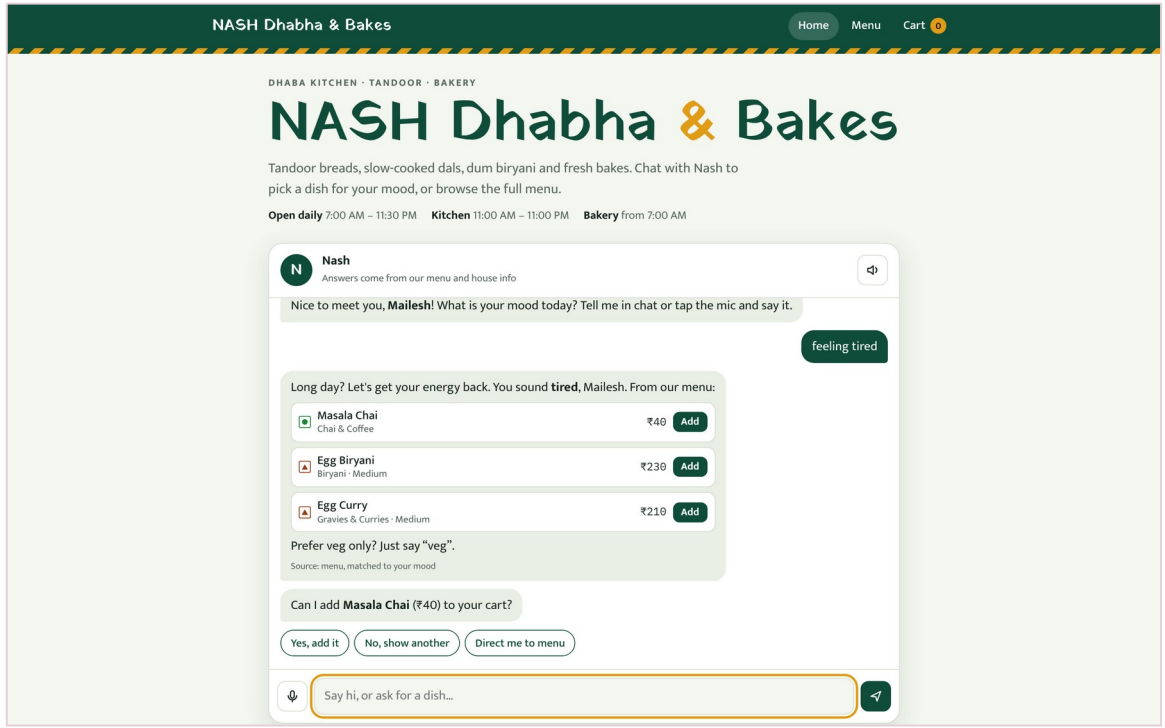
- Breads
- Rice
- Gravies & Curries
- Biryani
- Starters
- Juices & Lassi
- Chai & Coffee
- Desserts
- Ice Cream & Kulfi
- Bakes

Plus house-info pages: timings, payment, veg/non-veg kitchen, chef notes, spice, Jain & allergies, prep time, GST, about, ordering

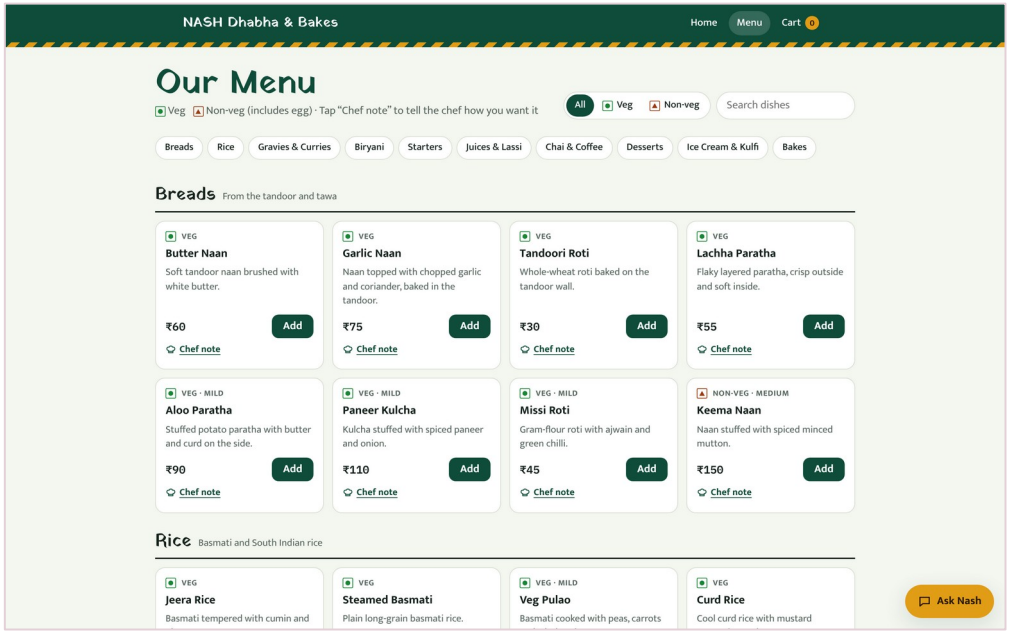
Planned fields for each dish

Field	Example
Name · category · price	Paneer Butter Masala · Curries · ₹260
Veg / non-veg / egg	Veg
Spice level	Mild
Description	Paneer in a creamy tomato-butter gravy
Mood tags	comfort, rich, celebrate
(Phase 2) texture, main ingredient, pairing	gravy · paneer · Butter Naan

Phase-1 Prototype (Working Demo)



v1 home: chat greets, asks name and mood, suggests dishes



v1 menu: 78 dishes, veg / non-veg, chef notes

Already working in Phase 1

Greeting → name → mood → suggestions · cart · chef notes · checkout · order updates · bill · payment · feedback

Planned Evaluation

Metric	How it will be measured	Target
Answer accuracy	Customer questions with rules checked against the menu	≥ 90%
Generalisation	Held-out questions written after tuning, run once	≥ 85%
Grounding	Count invented dishes / prices shown to customers	0
Task completion	End-to-end scenarios: name → order → bill → feedback	100% of steps
Robustness	Typos, Hinglish, XSS, prompt injection, server crash	No crash, safe output
Voice	Speech → text → action; notes reach the kitchen	All voice tests pass
Latency	Median reply time (engine and API)	< 1 s

Work Plan: Phase 1 → Phase 2

1

Phase 1

Prototype

Single-file app, menu, mood chat, order flow

2

Step 2

Backend

Express API, sessions, orders, kitchen display

3

Step 3

Waiter brain

Dish facts, request filters, chat-speak

4

Step 4

Voice

Speech-to-text chat, voice chef notes

5

Step 5

Gemini + guardrails

Grounded generation, validator, fallback

6

Step 6

Evaluate & share

Test suites, report, LinkedIn video

Phase 1 is complete (dark). Steps 2–6 are planned for the final review.

Risks & Mitigations

LLM invents dishes or prices

Mitigation: Retrieve first; validate every answer; rule-based fallback

Microphone blocked or unsupported

Mitigation: Clear message, typing always works, phone-recorder fallback

Hinglish, typos, chat-speak

Mitigation: Normalisation, spelling map, careful typo correction

No internet / API limits

Mitigation: Same engine runs offline in the browser; free-tier usage

Malicious input (XSS, injection)

Mitigation: Escape all output; model never controls cart or payment

Wrong diet advice

Mitigation: Explicit veg / non-veg / egg facts; never guess diet

Mapping to Course Outcomes

Course outcome (25ARE471)	How this project will address it
CO1 · Understand generative AI and its tools	RAG with vector similarity search, an LLM (Gemini), speech-to-text and text-to-speech
CO2 · AI principles and ethics	Grounded answers, validation, diet safety, privacy, honesty when a dish isn't available
CO3 · Apply GenAI tools to applications	A working restaurant assistant: Q&A, recommendations, voice notes, order automation
CO4 · Advanced GenAI concepts	Human-AI interaction design, AI-assisted coding, continuous evaluation (LLMOps)

Thank You!

Questions and feedback welcome

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